SIMATS ENGINEERING

ASSESSMENT TOOL 2 – Concept Mapping

COURSE NAME: Cryptography and Network Security
COURSE CODE: CSA5104

COURSE OUTCOME COVERED

CO1: Apply classical encryption techniques using symmetric and asymmetric ciphers to encrypt and decrypt data for the ensuring data security. (BL3, BL4)

Assessment Tool Weightage: 20%

QUESTIONS: 5 **Total Marks: 100**

Annexure A

CONCEPT MAPPING

Code of Conduct:

*I, **G. Padma Bhindya (192565049)** certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:



Annexure A

CONCEPT MAPPING

1. Create a concept map linking the following: Security Attacks, Threats, Risks and Security Goals (Confidentiality, Integrity, Availability. Perform the following tasks:

a. Show how each attack leads to specific threats and risks

b. Map how security goals help mitigate them Provide one real-world example for each link

2. Develop a concept map comparing substitution techniques: Caesar Cipher, Monoalphabetic Cipher, Vigenère Cipher and Hill Cipher. Perform the following tasks to connect based on: Key usage, Complexity and Security strength. Analyze which technique is most secure and why.

3. Construct a concept map for Classical Encryption Schemes: Substitution Techniques and Transposition Techniques. Perform the following tasks:

a. Show differences and similarities

i. Map how each affects the Data structure and Security level

ii. Analyze which is more vulnerable to attacks and why

4. Create a concept map connecting Steganography concepts: Data Hiding, Watermarking and Survivability. Perform the following tasks:

a. Show relationships between these concepts

b. Analyze how each contributes to secure communication

c. Include one real-world application for each

5. Draw a concept map linking number theory concepts used in cryptography: Modular Inverse, Extended Euclid Algorithm, Fermat's Little Theorem, Euler's Phi Function and Euler's Theorem. Perform the following tasks:

a. Show how each concept is mathematically connected

b. Map their role in encryption/decryption

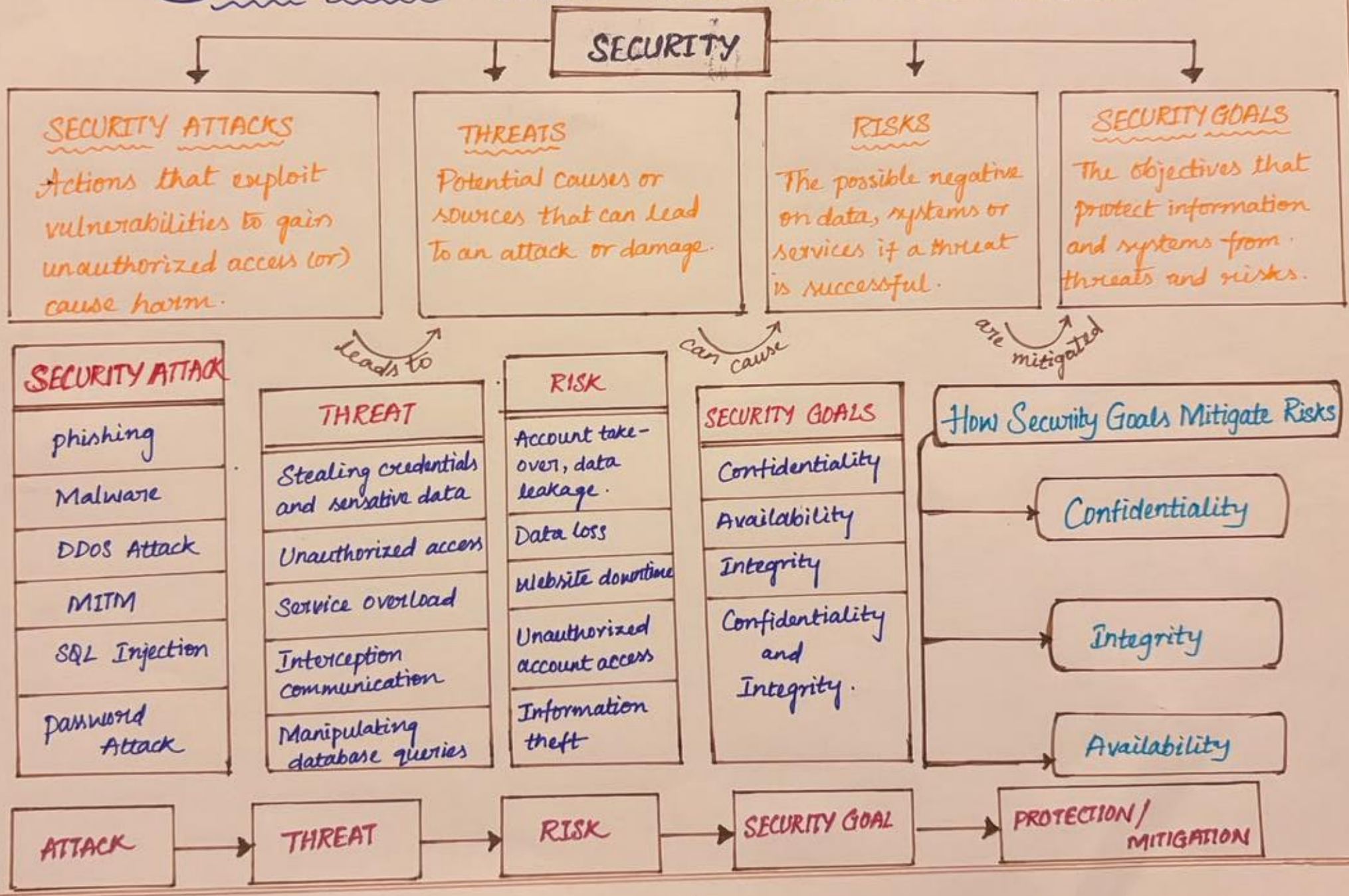
c. Analyze which concept is most critical for public key cryptography and justify

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Concepts	25	Demonstrates thorough understanding and effectively integrates multiple concepts/technologies	Good understanding with proper integration of most concepts	Basic understanding; limited or partial integration	Poor understanding; unable to integrate concepts
Experimental Design	30	Well-planned, systematic implementation with optimal solution	Correct implementation with minor issues	Basic implementation with errors or inefficiencies	Incorrect or incomplete implementation
Use of Tools	20	Effective and advanced use of tools/technologies with proper configuration	Appropriate use of tools with correct execution	Limited or inconsistent use of tools	Incorrect or minimal use of tools
Analysis of Results	15	Insightful analysis with accurate interpretation and validation of results	Good analysis with reasonable interpretation	Basic analysis with limited insights	Poor or incorrect analysis of results
Documentation	10	Clear, well-structured documentation with diagrams, code, and results	Organized documentation with necessary details	Basic documentation with missing details	Poor or incomplete documentation

CONCEPT MAP

SECURITY ATTACKS, THREATS, RISKS and SECURITY GOALS.



Number Theory Concepts Used In Cryptography

NUMBER THEORY

EULER's Theorem

If $\gcd(a,n)=1$, then $a^{\phi(n)} \equiv 1 \pmod{n}$.

[Generalizes Format's Little Theorem].

Example:
For $n=10$, $a=3$:
$3^4 \equiv 1 \pmod{10}$

Role in Cryptography:
Supports RSA's mathematical structure and correctness.

Modular Inverse

For $a \& m$, x is the modular inverse if, $a \times x \equiv 1 \pmod{m}$ (when $\gcd(a,m)=1$).

Example: $3 \times 7 \equiv 1 \pmod{10}$
→ 7 is the modular inverse of 3.

Role in cryptography:
Used in calculating private keys and decryption (e.g., RSA).

Extended Euclidean Algorithm

Find $\gcd(a,b)$ and coefficients x, y such that.
$ax + by = \gcd(a,b)$
Used to find modular inverse when $\gcd(a,m)=1$

Example: $\gcd(30,12)=6$
coefficient gives inverse when gcd = 1.

Role in cryptography:
Used in calculate the private exponent in RSA.

Euler's Phi Function

$\phi(n)$ gives the count of no. less than or equal to n that are relative to prime n.
$\phi(pq) = (p-1)(q-1)$

Example:
$\phi(15) = (3-1)(5-1)$
$= 2 \times 4 = 8$

Role in cryptography:
Used directly in RSA during key generation.

Format's Little Theorem

For prime p, if $\gcd(a=p)=1$, then $a^{(p-1)} \equiv 1 \pmod{p}$

Example:
$p=7$, $a=3$
$3^6 \equiv 1 \pmod{7}$

Role:
provides modular arithmetic principles, useful in cryptography algorithms.

Most Critical Concept

Euler's Phi Function ($\phi(n)$)
– it is directly used in RSA key generation, but all the concepts work together

How They Work Together in RSA

Euler's Phi Function ($\phi(n)$) → Extended Euclidean Algorithm → Modular Inverse (d) → Key Generation (p, q, n, e, d) → Encryption and Decryption ($C = M^e \bmod n$)